



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

Priyanka Paliwal  
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# Outline

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- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

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- Summary of methodologies
  - ☐ Data collection through API and Web Scraping
  - ☐ Data Wrangling
  - ☐ Exploratory Data Analysis using SQL, pandas and Matplotlib
  - ☐ Interactive Visual Analytics and Dashboard with Folium and Plotly Dash
  - ☐ Predictive Analysis
- Summary of all results
  - ☐ Best Hyperparameters for Logistic regression, SVM, Decision Tree Classifier and KNN Classifier
  - ☐ Method that performs best using test data.

# Introduction

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- Project background and context

SpaceX advertises Falcon 9 rocket launches on its website at a cost of \$62 million, while other providers charge upwards of \$165 million per launch. Much of SpaceX's savings come from the ability to reuse the first stage of their rockets. Therefore, if we can predict whether the first stage will land successfully, we can estimate the cost of a launch. This information is valuable for an alternate company, SpaceY, which may want to bid against SpaceX for rocket launches. The goal of this project is to create a machine learning pipeline to predict the successful landing of the first stage.

- Problems you want to find answers

- ☐ What factors determine if the rocket will land successfully?
- ☐ The interaction amongst various features that determine the success rate of a successful landing.
- ☐ What operating conditions needs to be in place to ensure a successful landing program.



Section 1

# Methodology

# Methodology

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## Executive Summary

- Data collection methodology:
  - The data was gathered from the SpaceX REST API and web scraping from Wikipedia.
- Perform data wrangling
  - The data collected is in form of a JSON object and HTML tables, after that data is converted into Pandas dataframe for visualization and analysis.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Use of machine learning to determine if the first stage of Falcon 9 will land successfully.

# Data Collection

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- The Data was collected using two methods in the following way:

- ☐ Data collection was done using SpaceX REST API

- ☐ Then the response content was converted to json using `json()` function call and turn it into pandas dataframe using `.json_normalize()`.



- ☐ Data was also collected using web scraping from Wikipedia for Falcon 9 launch records with BeautifulSoup

- ☐ The launch records was extracted as HTML table, parse the table and convert it to a pandas dataframe.



# Data Collection – SpaceX API

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The process for data collection using SpaceX REST API is as follows:



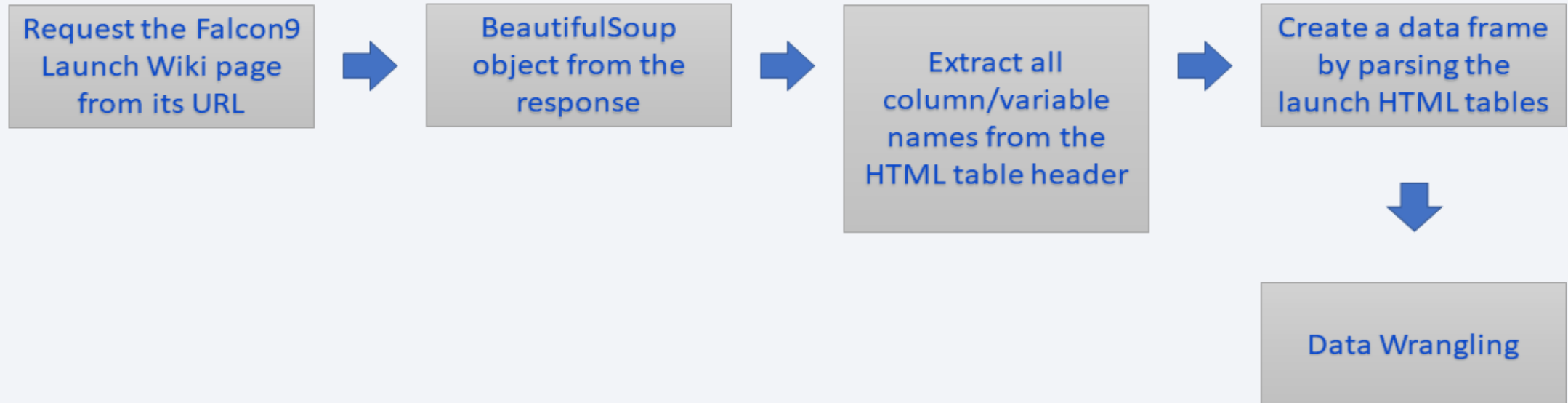
The GitHub URL: [Data Collection using SpaceX REST API](#)



# Data Collection - Scrapping

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The process of data collection using Web scrapping is as follows:

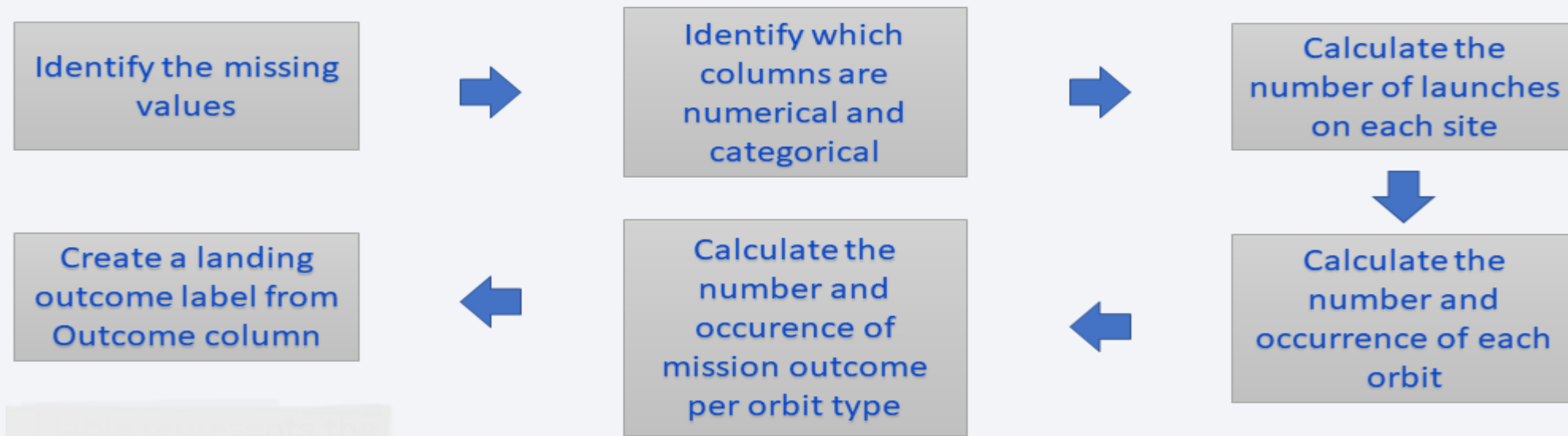


The GitHub URL : [Data Collection using Web Scrapping](#)

# Data Wrangling

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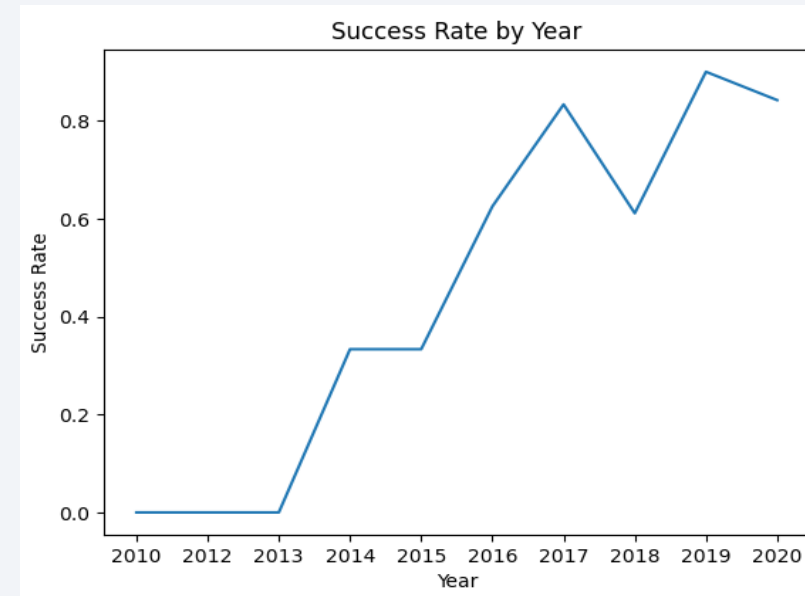
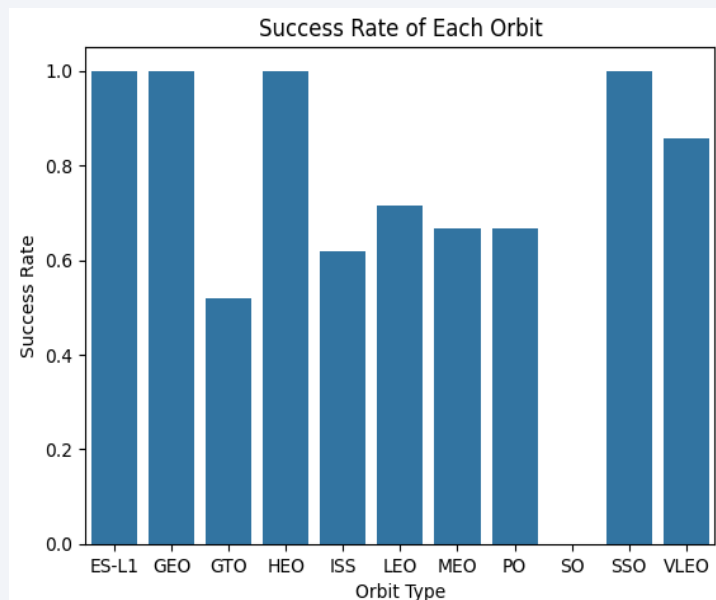
The data was processed by cleaning it, identifying the missing values, identify the categorical columns and converting it into numerical, calculating the no. of launches on each site, calculating the no. and occurrence of each orbit and so on. The process is shown below:



The GitHub URL : [Data Wrangling](#)

# EDA with Data Visualization

The data was explored by visualizing various plots such as Catplot showing the relationship between Flight no., Payload, launch site, orbit type. Bar chart visualizing the success rate of each orbit type and line chart to visualize the launch success yearly trend.



GitHub URL : [EDA with Data Visualization](#)

# EDA with SQL

- The SQL queries used for EDA are as follows:

- ❑ Display the names of the unique launch sites in the space mission: `select DISTINCT Launch_Site from SPACEXTABLE`
- ❑ Display 5 records where launch sites begin with the string 'CCA' : `select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5`
- ❑ Display the total payload mass carried by boosters launched by NASA (CRS): `select sum(PAYLOAD_MASS__KG_) AS Total_Payload_Mass from SPACEXTABLE where Customer = 'NASA (CRS)'`
- ❑ Display average payload mass carried by booster version F9 v1.1: `select avg(PAYLOAD_MASS__KG_) AS Average_Payload_Mass from SPACEXTABLE where Booster_Version LIKE 'F9 v1.1%'`
- ❑ List the date when the first succesful landing outcome in ground pad was acheived. : `select min(Date) from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)'`
- ❑ List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000: `select Booster_Version from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000`
- ❑ List the total number of successful and failure mission outcomes: `select Mission_Outcome, Count(*) from SPACEXTABLE group by Mission_Outcome`
- ❑ List all the booster\_versions that have carried the maximum payload mass. Use a subquery.: `select Booster_Version from SPACEXTABLE where PAYLOAD_MASS__KG_ = (select max(PAYLOAD_MASS__KG_) from SPACEXTABLE)`
- ❑ List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.: `SELECT CASE WHEN substr("Date", 6, 2) = '01' THEN 'January' WHEN substr("Date", 6, 2) = '02' THEN 'February' WHEN substr("Date", 6, 2) = '03' THEN 'March' WHEN substr("Date", 6, 2) = '04' THEN 'April' WHEN substr("Date", 6, 2) = '05' THEN 'May' WHEN substr("Date", 6, 2) = '06' THEN 'June' WHEN substr("Date", 6, 2) = '07' THEN 'July' WHEN substr("Date", 6, 2) = '08' THEN 'August' WHEN substr("Date", 6, 2) = '09' THEN 'September' WHEN substr("Date", 6, 2) = '10' THEN 'October' WHEN substr("Date", 6, 2) = '11' THEN 'November' WHEN substr("Date", 6, 2) = '12' THEN 'December' END AS month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE WHERE substr("Date", 0, 5) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)'`
- ❑ Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.: `SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY outcome_count DESC`

- GitHub URL : [EDA with SQL](#)

# Build an Interactive Map with Folium

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- The map objects such as markers, circles, lines, etc. were created and added to a folium map
  - ❑ folium.Circle and folium.Marker to add a highlighted circle are with text label on a specific coordinate for each launch site on the site map.
  - ❑ MarkerCluster object for simplify a map containing many markers having the same coordinate.
  - ❑ MousePosition on the map to get coordinate for a mouse over a point on the map.
  - ❑ folium.Polyline object to draw a line between a launch site to its closest city, railway and highway.
- GitHub URL : [Interactive Map with Folium](#)



# Build a Dashboard with Plotly Dash

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- The interactive dashboard was created with Plotly Dash. It contains input components such as dropdown list and range slider to interact with pie chart and scatter point chart.
  - ❑ A launch site dropdown input component: There are 4 different launch sites, and a dropdown menu let us select different launch site
  - ❑ A callback function to render success pie chart based on selected site dropdown
  - ❑ A range slider to select payload: the slider is to be able to easily select different payload range and see if we can identify some visual patterns.
  - ❑ A callback function to render the success payload scatter chart to visually observe how payload may be correlated with mission outcomes for selected sites.
- GitHub URL : [Dashboard with Plotly Dash](#)

# Predictive Analysis (Classification)

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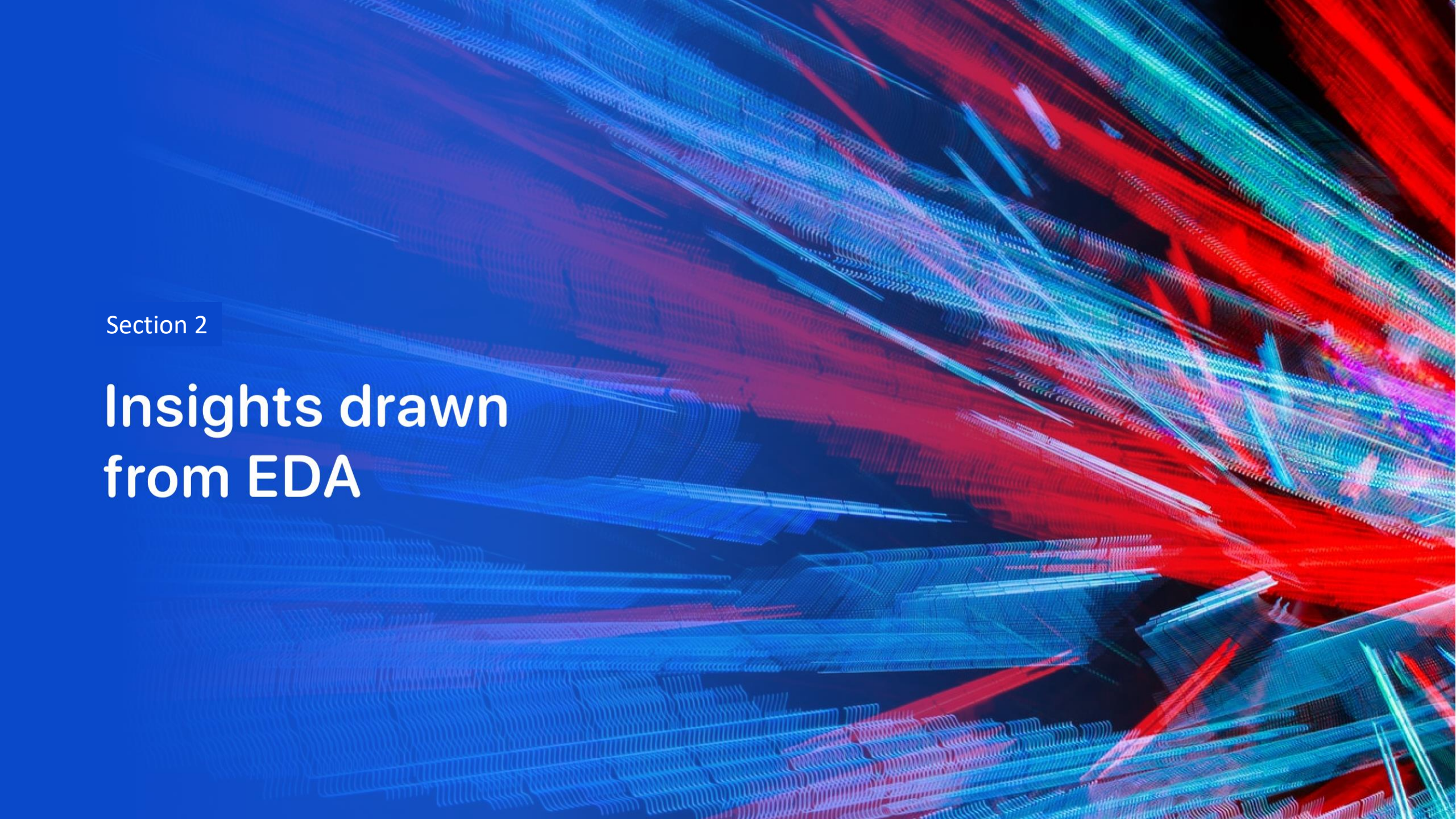
- We loaded the data using numpy and pandas, transformed the data, split the data into training and testing.
- Built different machine learning models like Logistic Regression, SVM, Decision Tree Classifier and KNN classifier and tune different hyperparameters using GridSearchCV.
- Found the accuracy for each model and found the model with best performance on test data.
- GitHub URL : [Predictive Analysis](#)

# Results

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- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



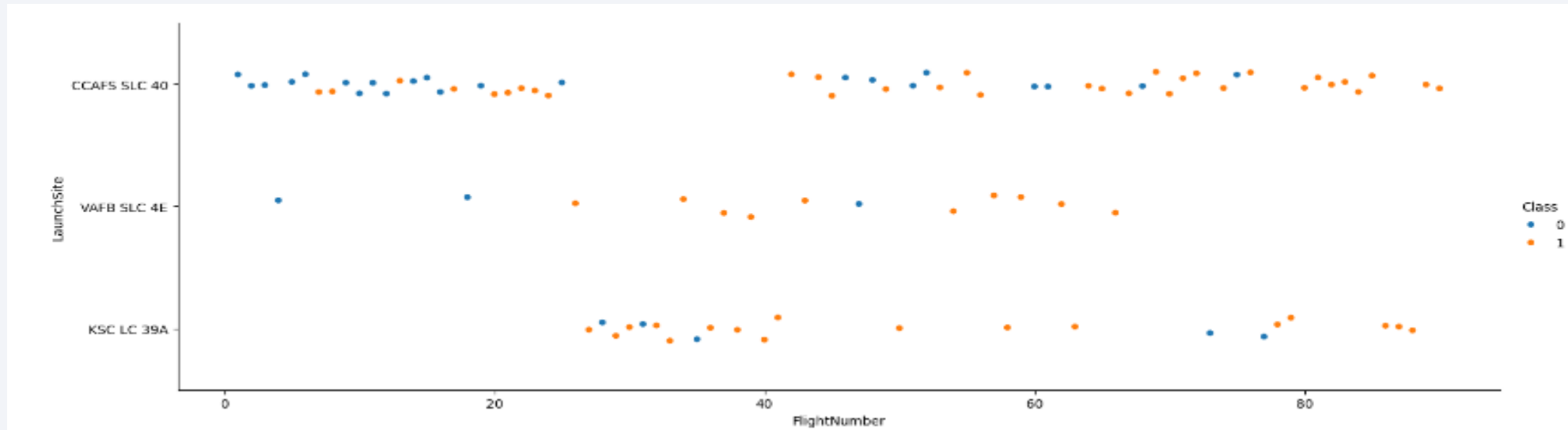
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

# Insights drawn from EDA

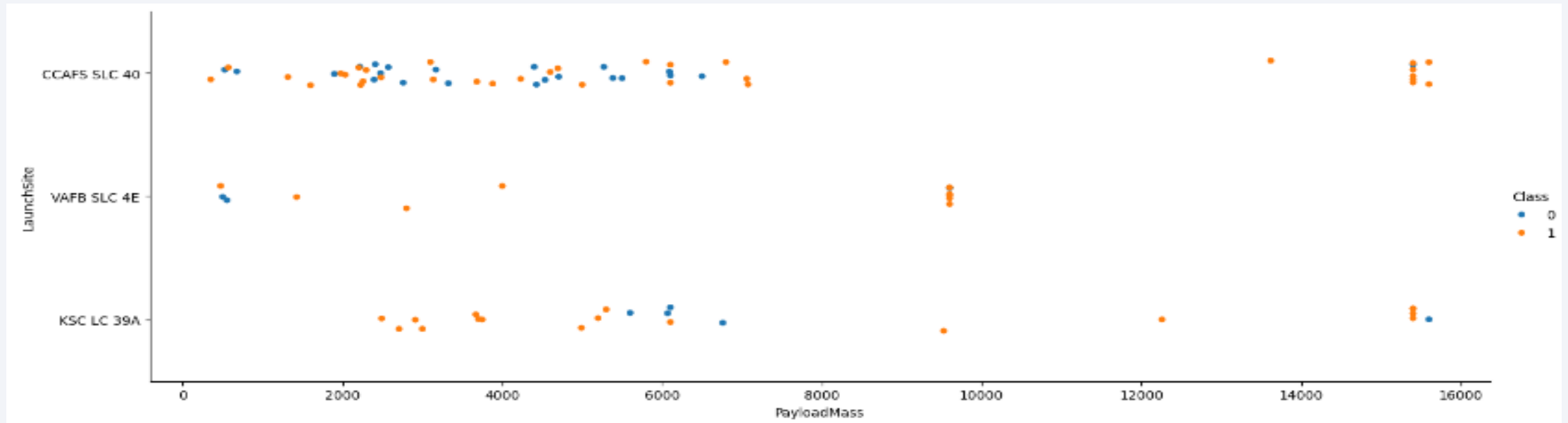


# Flight Number vs. Launch Site



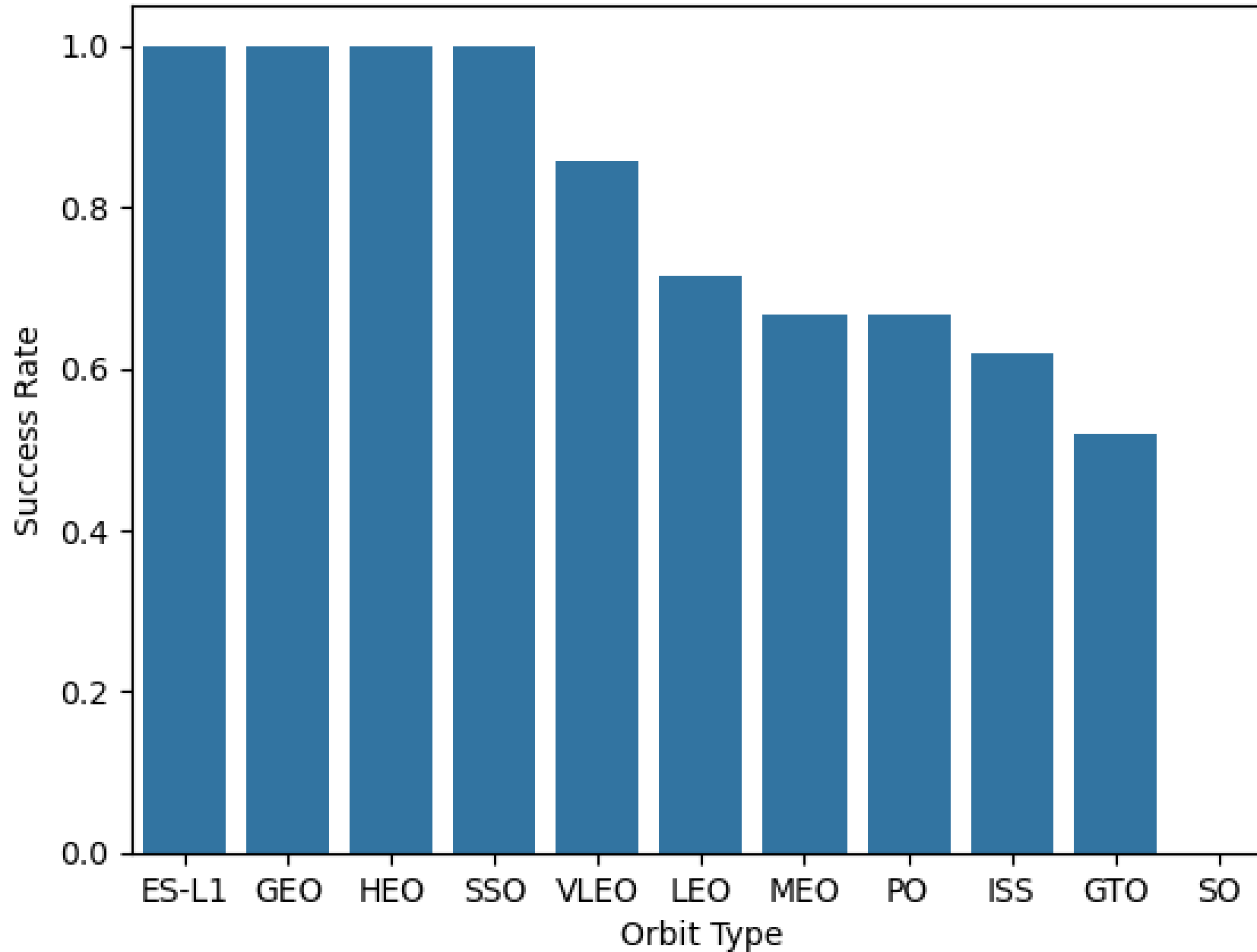
- For every Launch Site, successful rate is increasing with time, especially for CCAFS SLC 40.
- VAFB SLC 4E and KSC LC 39A has a higher successful rate but represents one third of the total launches.

# Payload vs. Launch Site



- No rocket is launched for heavy payloads at VAFB SLC 4E Launch Site.
- For CCAFS SLC 40, rockets are launched for below 8000 kg and more than 13000Kg Payload mass but not in between.
- For KSC SLC 39A, no rockets are launched for lower payload mass

Success Rate of Each Orbit

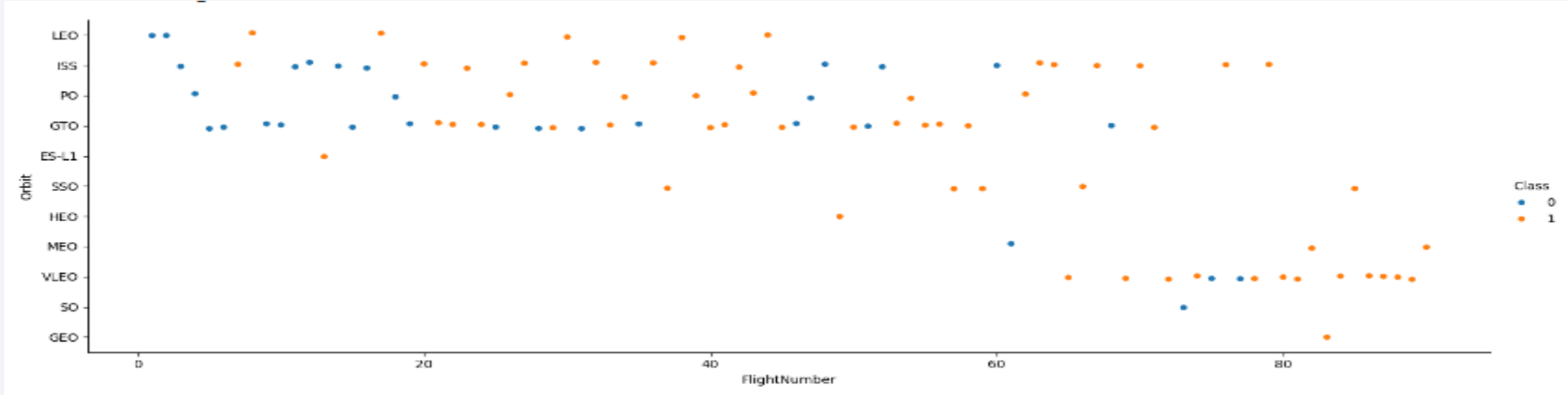


## Success Rate vs. Orbit Type

- The first 4 Orbit types has the best successful rate.

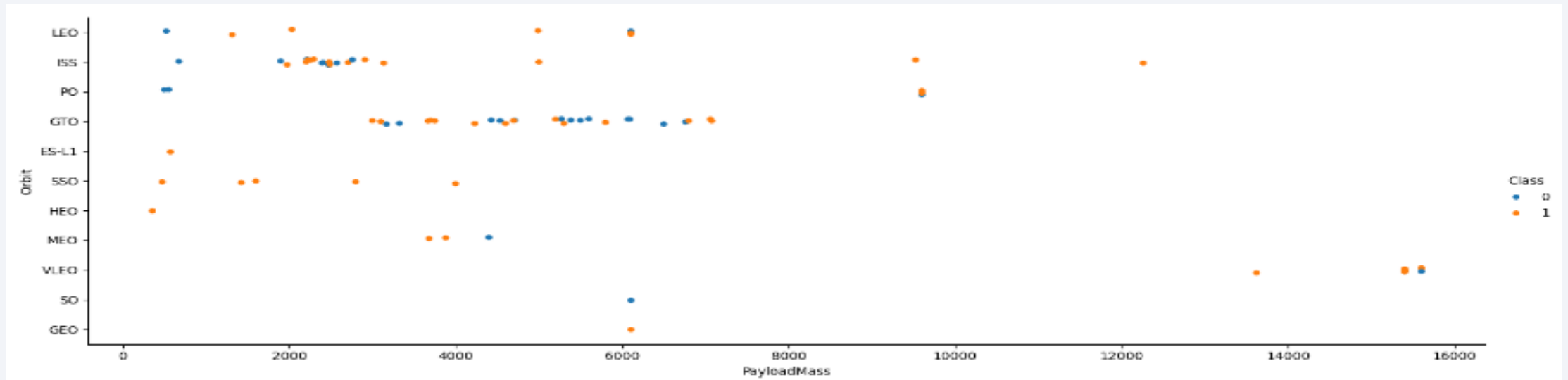


# Flight Number vs. Orbit Type



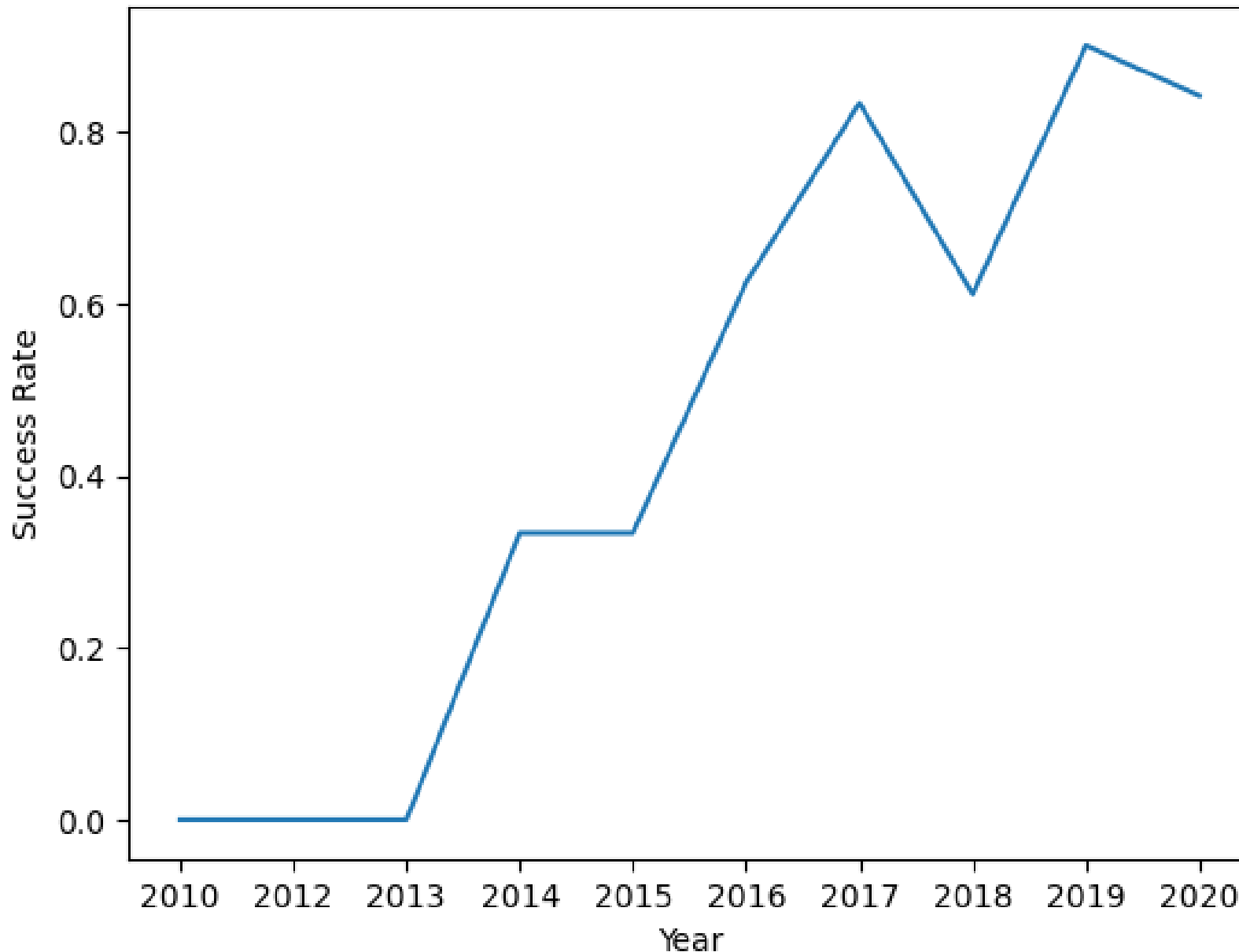
- There are more failures at the beginning of the series of launches, but after the first 40 launches, the ratio improves by reducing the 50% of unsuccessful landings.
- GTO and ISS orbits has the higher concentration of launches with the lowest ratio of successful landings.

# Payload vs. Orbit Type



- With heavy payloads the successful landing rate are more for Polar, LEO and ISS
- For GTO, we cannot distinguish this well as both, positive landing rate and negative landing are both there.

Success Rate by Year



## Launch Success Yearly Trend

- The success rate since 2014 kept increasing until 2020.

Display the names of the unique launch sites in th

]:

```
%sql select DISTINCT Launch_Site from SPACE
```

```
* sqlite:///my_data1.db
```

Done.

]:

**Launch\_Site**

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

# All Launch Site Names

- The names of the unique launch sites are:
- CCAFS LC – 40
- VFAF SLC - 4E
- KSC LC – 39A
- CCAFS SLC - 40

```
%sql select * from SPACEXTABLE where Launch_Site like 'CCA%' limit 5
```

```
sqlite:///my_data1.db  
ne.
```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       | PAYLOAD_MASS_KG_ | Orbit     | Cu |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|------------------|-----------|----|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFS LC-40 | Dragon Spacecraft Qualification Unit                          | 0                | LEO       |    |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0                | LEO (ISS) |    |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFS LC-40 | Dragon demo flight C2                                         | 525              | LEO (ISS) |    |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFS LC-40 | SpaceX CRS-1                                                  | 500              | LEO (ISS) |    |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFS LC-40 | SpaceX CRS-2                                                  | 677              | LEO (ISS) |    |

# Launch Site Names Begin with 'CCA'

5 records where launch sites begin with 'CCA'. The query uses WHERE, LIKE and LIMIT.

# Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

In [42]:

```
%sql select sum(PAYLOAD_MASS_KG_) AS Total_Payload_Mass from SPACEXTABLE where Customer = 'NASA (CRS)'
```

\* sqlite:///my\_data1.db  
Done.

Out[42]: Total\_Payload\_Mass

45596

The total payload carried by boosters from NASA is 45596 Kg



# Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

In [43]:

```
%sql select avg(PAYLOAD_MASS_KG_) AS Average_Payload_Mass from SPACEXTABLE where Booster_Version LIKE 'F9 v1.1%'
```

```
* sqlite:///my_data1.db  
Done.
```

Out[43]:

| Average_Payload_Mass |
|----------------------|
|----------------------|

|                    |
|--------------------|
| 2534.6666666666665 |
|--------------------|

The average payload mass carried by booster version F9 v1.1 is 2534.665 Kg

# First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint: Use min function*

In [48]:

```
%sql select min(Date) from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db  
Done.
```

Out[48]:

| <b>min(Date)</b> |
|------------------|
| 2015-12-22       |

The first successful landing outcome on ground pad is on 22 Dec 2015

## Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [51]: %sql select Booster_Version from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000
```

\* sqlite:///my\_data1.db  
Done.

Out[51]:

| Booster_Version |
|-----------------|
| F9 FT B1022     |
| F9 FT B1026     |
| F9 FT B1021.2   |
| F9 FT B1031.2   |

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:
  - ☐ F9 FT B1022
  - ☐ F9 FT B1026
  - ☐ F9 FT B1021.2
  - ☐ F9 FT B1031.2

# Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
In [53]: %sql select Mission_Outcome, Count(*) from SPACEXTABLE group by Mission_Outcome
```

\* sqlite:///my\_data1.db  
Done.

Out[53]:

| Mission_Outcome                  | Count(*) |
|----------------------------------|----------|
| Failure (in flight)              | 1        |
| Success                          | 98       |
| Success                          | 1        |
| Success (payload status unclear) | 1        |

The total number of successful and failure mission outcomes are 99 and 1 respectively.

List all the booster\_versions that have carried the maximum payload mass. Use

In [62]:

```
%sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_
```

```
* sqlite:///my_data1.db  
Done.
```

Out[62]:

| Booster_Version |
|-----------------|
| F9 B5 B1048.4   |
| F9 B5 B1049.4   |
| F9 B5 B1051.3   |
| F9 B5 B1056.4   |
| F9 B5 B1048.5   |
| F9 B5 B1051.4   |
| F9 B5 B1049.5   |
| F9 B5 B1060.2   |
| F9 B5 B1058.3   |
| F9 B5 B1051.6   |
| F9 B5 B1060.3   |
| F9 B5 B1049.7   |

# Boosters Carried Maximum Payload

The names of the booster which have carried the maximum payload mass are shown here

# 2015 Launch Records

List the records which will display the month names, failure landing\_outcomes in drone ship ,booster versions, launch\_site for the months in year 2015.

**Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.**

In [67]:

```
%sql SELECT CASE WHEN substr("Date", 6, 2) = '01' THEN 'January' WHEN substr("Date", 6, 2) = '02' THEN 'February' WHEN subs
```

\* sqlite:///my\_data1.db  
Done.

Out[67]:

| month   | Landing_Outcome      | Booster_Version | Launch_Site |
|---------|----------------------|-----------------|-------------|
| January | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40 |
| April   | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40 |

- The failed landing\_outcomes in drone ship, their booster versions, and launch site names for in year 2015

## Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
In [68]: %sql SELECT "Landing_Outcome", COUNT(*) AS outcome_count FROM SPACEXTABLE WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
```

\* sqlite:///my\_data1.db  
Done.

Out[68]:

| Landing_Outcome        | outcome_count |
|------------------------|---------------|
| No attempt             | 10            |
| Success (drone ship)   | 5             |
| Failure (drone ship)   | 5             |
| Success (ground pad)   | 3             |
| Controlled (ocean)     | 3             |
| Uncontrolled (ocean)   | 2             |
| Failure (parachute)    | 2             |
| Precluded (drone ship) | 1             |

- The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, is ranked in descending order



A satellite view of Earth from space, showing the curvature of the planet and the glowing lights of cities and continents against the dark background of space. The Earth's surface is a mix of dark blue oceans and lighter blue/white landmasses, with numerous bright yellow and orange lights indicating urban areas.

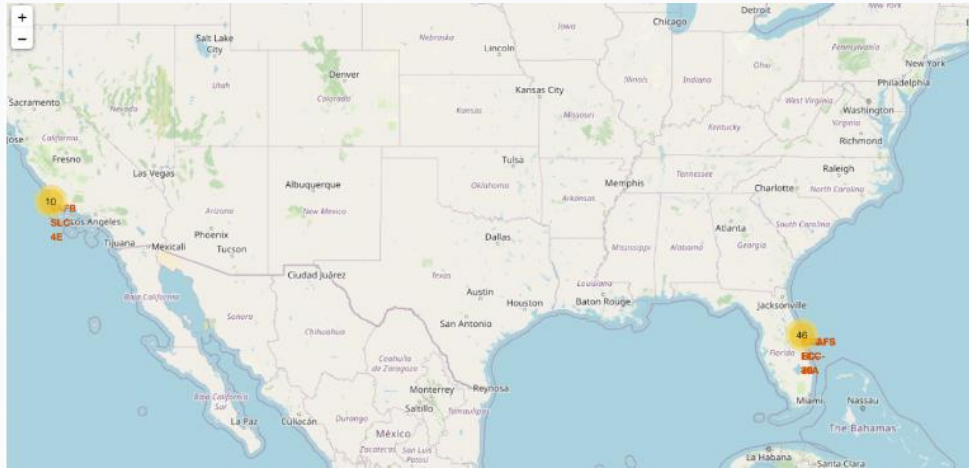
Section 3

# Launch Sites Proximities Analysis



# All Launch Sites

- All Launch Sites are in close proximity to the coast and in restricted area.

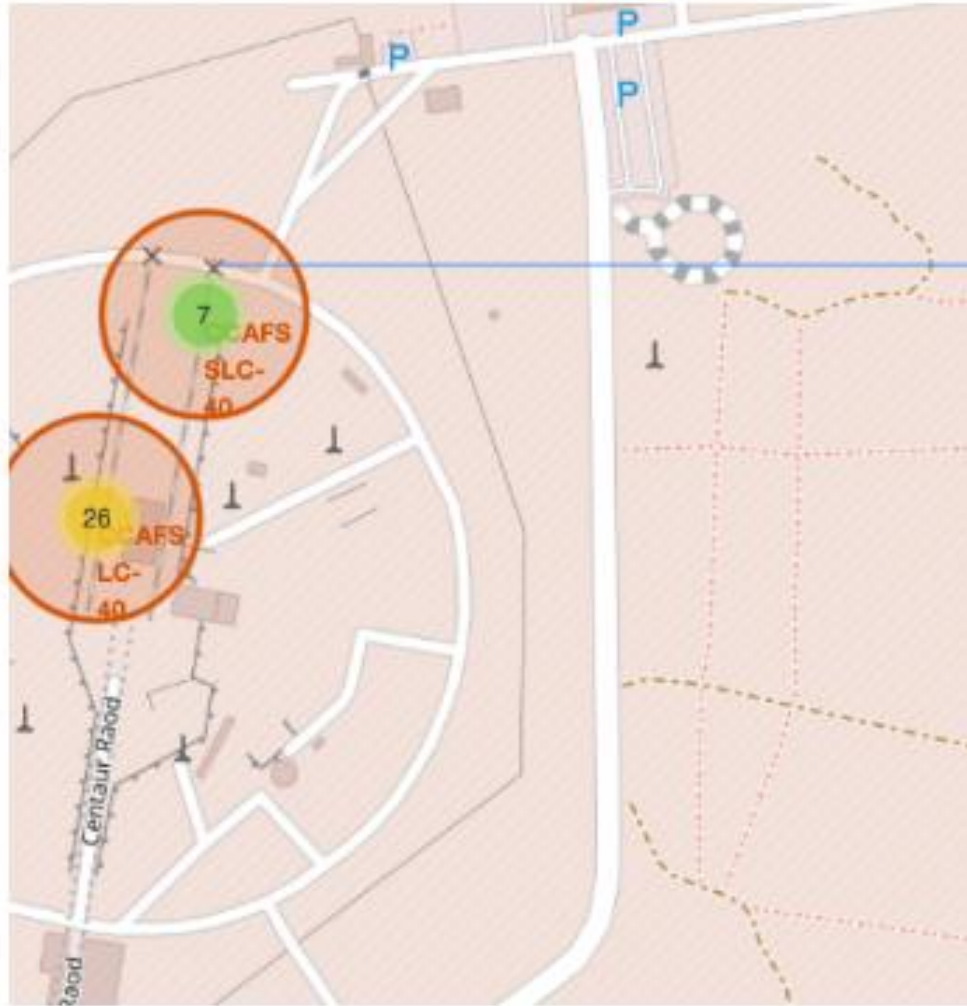


# The success/failed launches for each site on the map

The first map shows clusters for every launch site, the second shows a green marker if a launch site was successful and red marker if a lunch was failed.

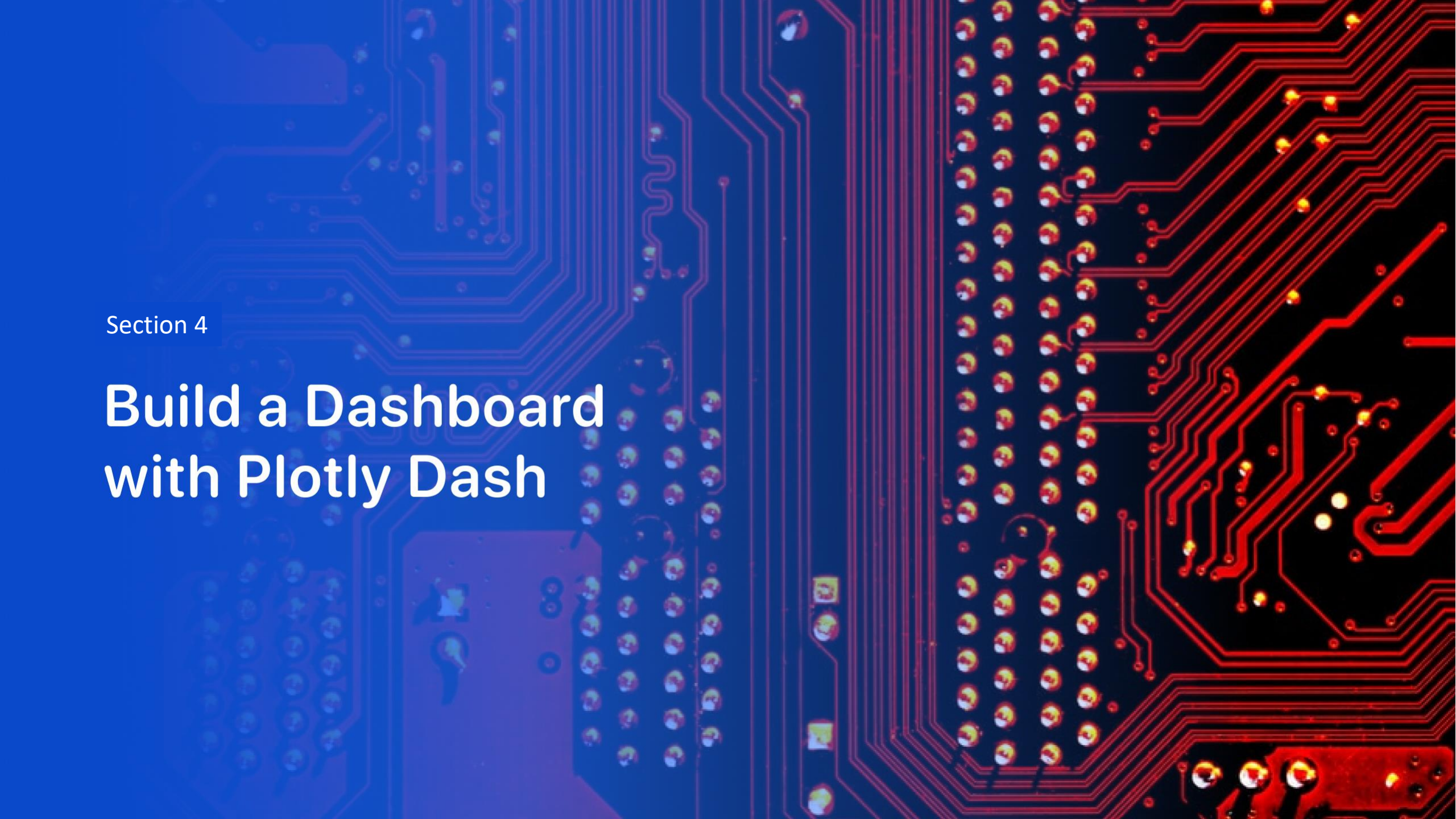






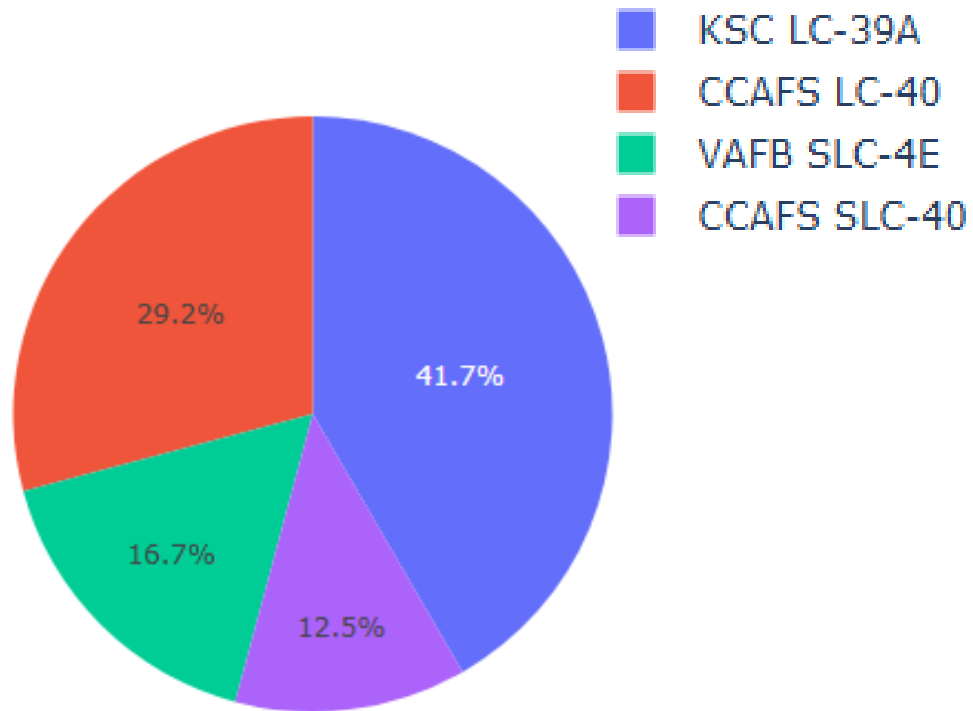
# A Launch Site And Its Proximities

Launch Sites are near to railways,  
roads, highways and coastlines.

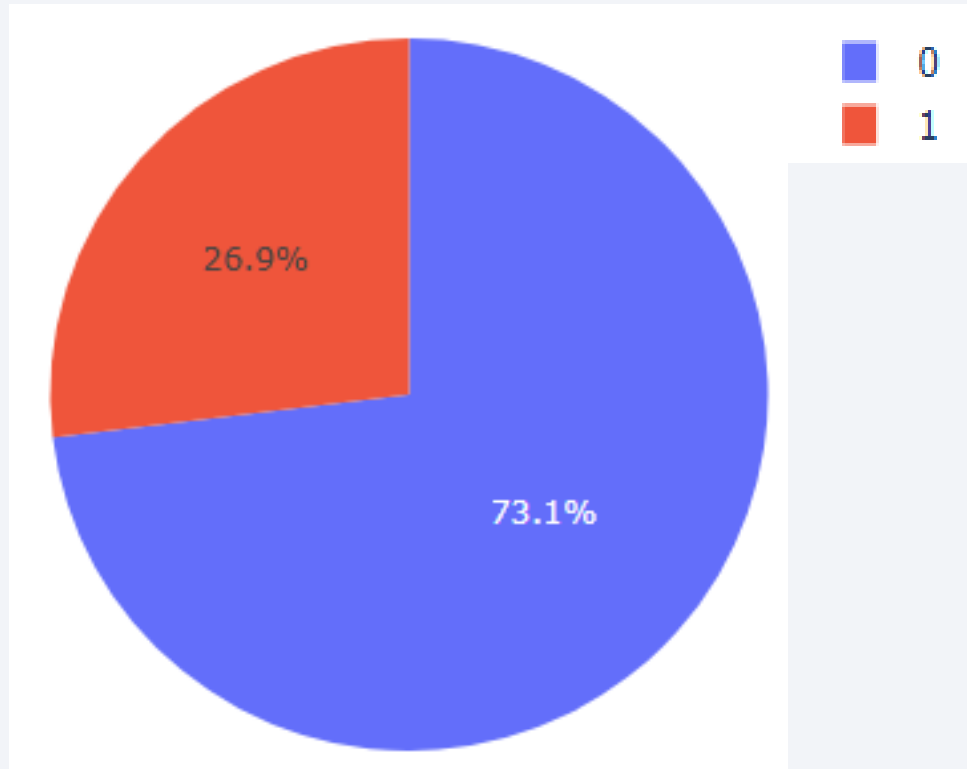


Section 4

# Build a Dashboard with Plotly Dash



Pie Chart showing  
the success  
percentage achieved  
by each Launch Site

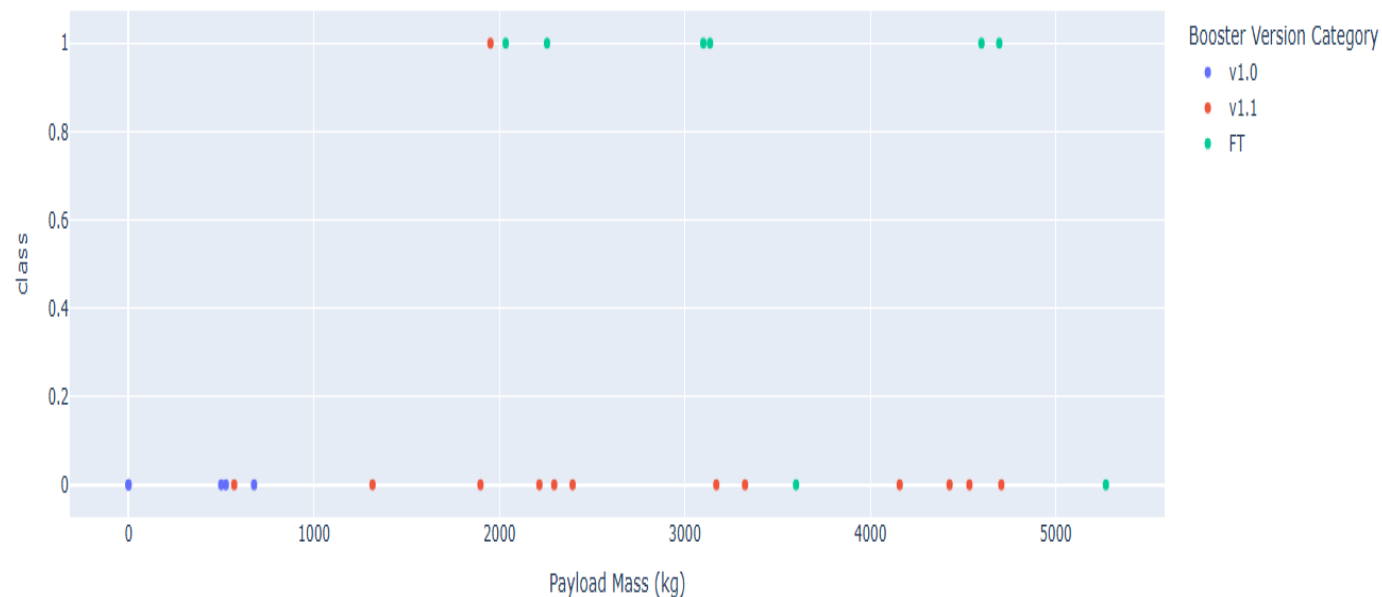


Pie Chart showing the Launch Site with highest launch Success ratio

Payload range (Kg):



Correlation between Payload and Success for CCAFS LC-40



# Correlation between Payload and Success from CCAFS LC - 40

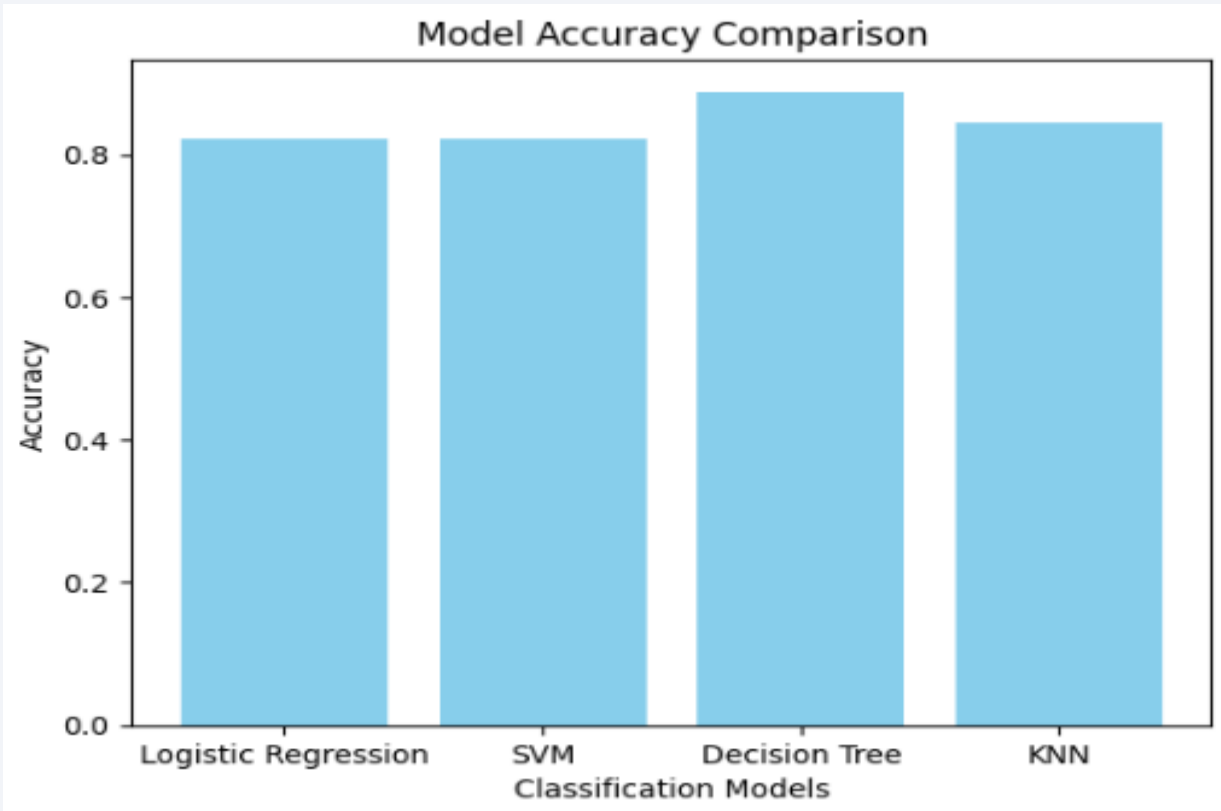


Section 5

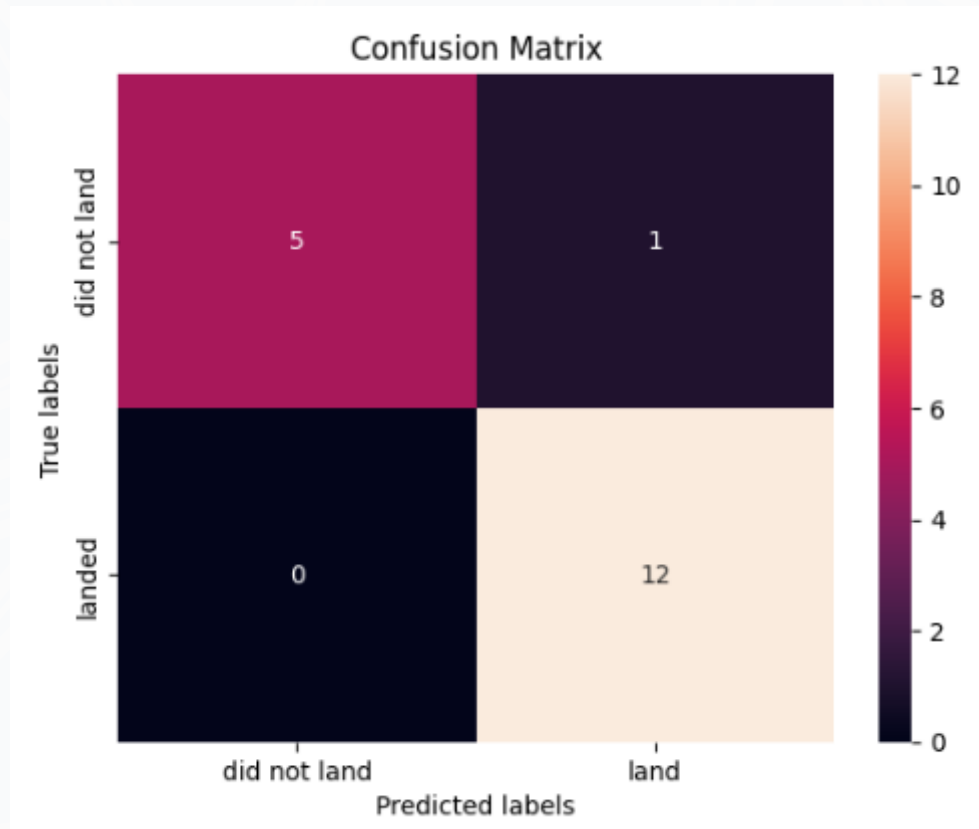
# Predictive Analysis (Classification)

# Classification Accuracy

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- The built model accuracy for all built classification models, is visualized in a bar chart
- Decision tree has the highest classification accuracy



# Confusion Matrix

The confusion matrix of the best performing model i.e. Decision Tree Classifier is shown here.

There is only 1 false positive for this model

# Conclusions

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We can conclude that:

- The larger the flight amount at a launch site, the greater the success rate at a launch site.
- Launch success rate started to increase in 2013 till 2020.
- Orbits ES-L1, GEO, HEO, SSO, VLEO had the most success rate.
- KSC LC-39A had the most successful launches of any sites.
- The Decision tree classifier is the best machine learning algorithm for this task.

# Appendix

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- For Python code snippets, SQL queries, charts, Notebook outputs, the Git hub link for the repository is as follow:

[Data Science Capstone](#)



Thank you!

